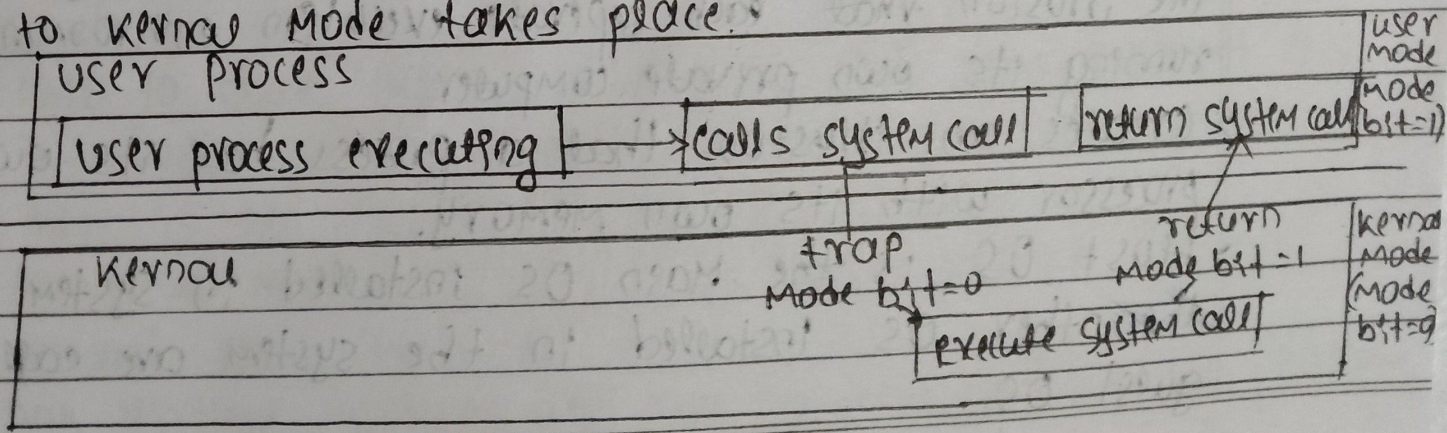
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02) Define operating systems. Explain the dual-mode operating system with a neat diagram

→ An operating system is a system software that acts as as intermediary between a user of a computer and the computer hardware.

Dual Mode Operation

- Since the operating system and the user programs share the hardware and software resources of the computer system, it has to be made sure that an error in a user program cannot cause problems to other programs and the OS running in the system.
- The approach taken is to use a hardware support that allows us to differentiate among various modes of execution.
- The system can be assumed to work in two seprate modes of operation: i) user mode ii) kernal mode.
- A hardware bit of the computer, called the mode bit, is used to indicate the current mode: kernal (0) or user(1). With the mode bit, we are able to distinguish b/w a task that is executed by the OS & one that is executed by user.
- When the computer system is executing a user application, the system is in user mode. When a user application requests a service from the O.S, the transition from user to kernal mode takes place.

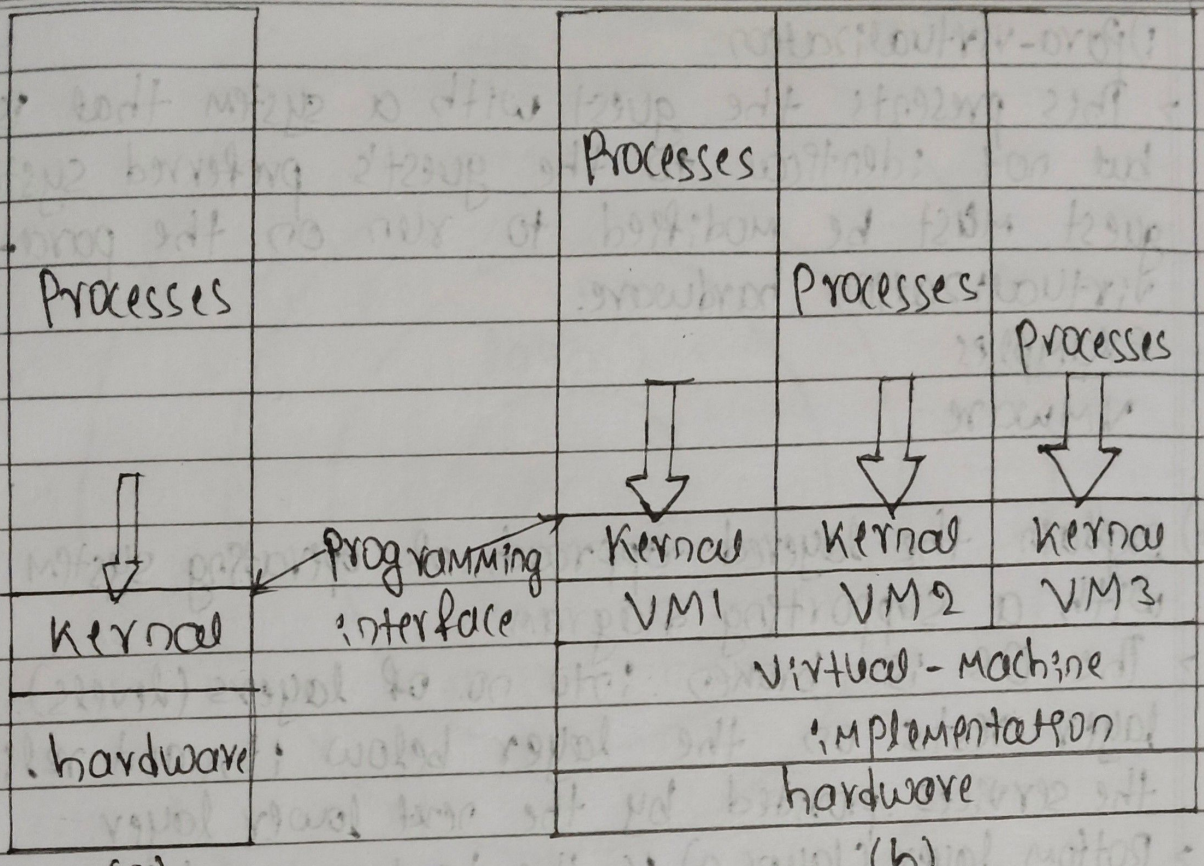


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→ At system boot time, the hardware starts in kernal mode. The O.S is then loaded and starts user applications in uses mode. Whenever a trap or interrupt occurs, the hardware switches from user mode to kernal mode
→ The dual mode of operation provides us with the means for protecting the O.S from errant users - and errant users from one another.
→ The hardware allows privilaged instructions to be executed only in kernal mode. If an attempt is made to execute a privilaged instruction in user mode, the hardware does not execute the instruction but rather treats it as illigal and traps it to the O.S.
→ Initial control is within the O.S, where instructions are executed in kernal mode. When control is given to a user application, the mode is set to user mode. Eventually, control is switched back to the O.S via an interrupt, a trap, or a system call.

03) With a neat diagram, explain the concept of the virtual machine.

→ The fundamental idea behind a virtual machine is to abstract the hardware of a single computer into several different execution environments, thereby creating the illusion that each seperate execution environment is running its own private computer
→ Creates an illusion that a process has its own processor with its own memory.
→ Host OS is the main OS installed in system and the other OS installed in the system are called guest OS.



System Modes. (a) Non-Virtual Machine (b) Virtual Machine

Implementation

- The virtual machine concept is useful, it is difficult to implement.
- Work is required to provide an exact duplicate of the underlying machine. The machine has 2 modes. User & Kernal mode.
- The virtual-machine software can run in Kernal mode, since it is the O.S system. It can execute in only user-mode.

Benefits

- Able to share the same hardware but runs different O.S.
- The host system is protected from the virtual machine.
- There is no direct sharing of resources.
- A virtual machine system is a perfect vechile for O.S research and development.
- Any virtual machines will run on any virtulization platform.

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i) Para-virtualization

→ This presents the guest with a system that is similar but not identical to the guest's preferred system. The guest must be modified to run on the para-virtualization hardware.

Examples

VMware

04) Explain the layered approach of operating system structure with a supporting diagram.

→ → The OS is broken into no. of layers (levels). Each layer rests on the layer below it, and relies on the services provided by the next lower layer.

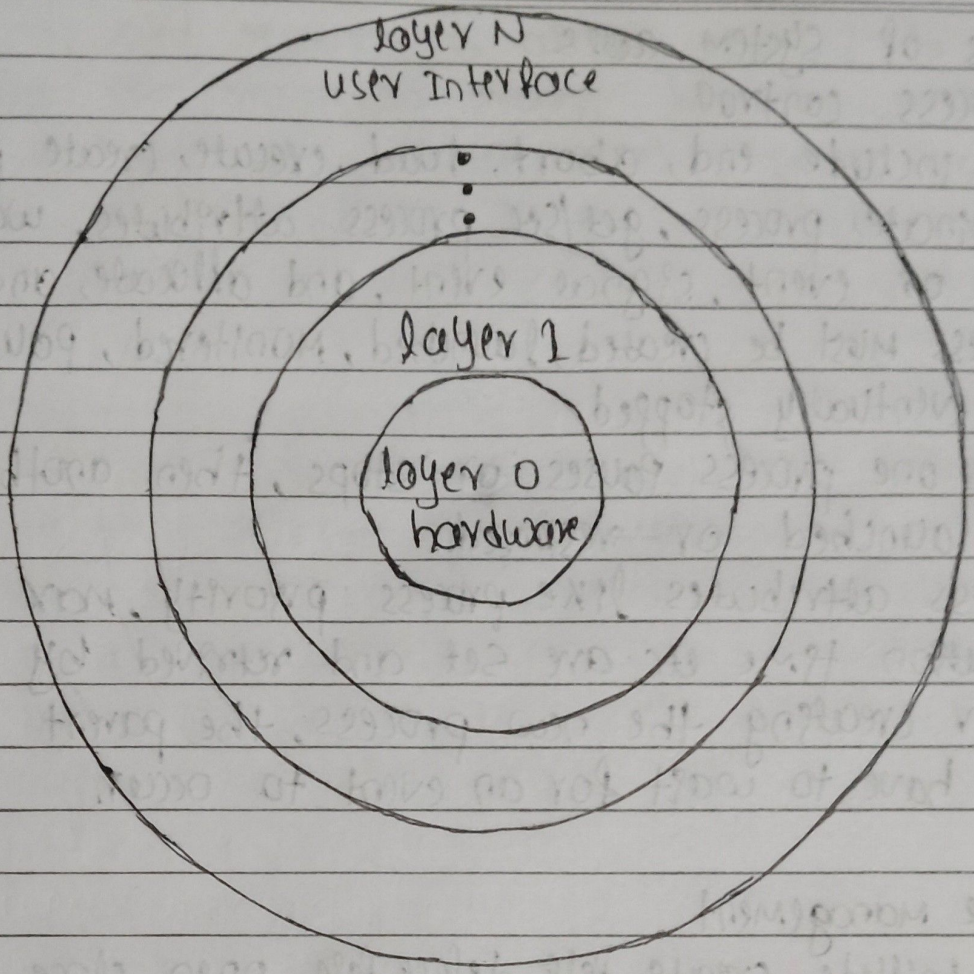
→ Bottom layer (layer 0) is the hardware and the topmost layer is the user interface.

→ A typical layer, consists of data structure and routines that can be invoked by higher-level layer.

→ Advantage of layered approach is simplicity of construction and debugging.

→ The layers are selected so that each uses functions and services of only lower-level layers. So simplifies debugging and system verification. The layers are debugged one by one, from the lowest and if any layer doesn't work, then errors is due to that layer only, as the lower layers are already debugged. Thus, the design & implementation are simplified.

→ A layer need not know how its lower-level layers are implemented. Thus hides the operations from higher layers.



A layered operating system

Disadvantages

> The various layers must be appropriately defined, as a layer can use only lower-level layers.
> Less efficient than other types, because any interaction with layer 0 required from top layer. The system call should pass through all the layers and penalty to layer 0. This is an overhead.

Q5) What are system calls? Briefly point out its type with illustrations.

→ > System calls provides an interface to the services of the O.S. These are generally written in C or C++, although some are written in assembly for optimal performance.

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Types of system calls

i) Process control

> It include end, abort, load, execute, create process, terminate process, get/set process attributes, wait for time or event, signal event, and allocate, and free memory.

-> Process must be created, launched, monitored, paused, resumed, and eventually stopped.

> When one process pauses or stops, then another must be launched or resumed.

> Process attributes like process priority, max allowable execution time etc are set and retrived by OS

> After creating the new process, the parent process may have to wait for an event to occur.

ii) File Management

> It include create file, delete file, open, close, read, write, reposition, get file attributes and set file attributes.

> After creating a file, the file is opened. Data is read or written to a file.

> The file pointer may need to be repositioned to a point.

> The file attributes like filename, filetype, permissions etc are set and retrived using system calls.

> These operations may also be supported for directories as well as ordinary files.

iii) Device Management.

> It include request device, release device, read, write, reposition, get/set device attributes & logically attach or detach devices.

> When a process needs a resource, a request for resource is done.

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> If requested resource is already attached to some other process, the requesting process has to wait.
> In multiprogramming system, after a process uses the device, it has to returned to OS, so that another process can use the device. Devices may be physical, or virtual/abstract.

iv) Information maintaince
> It includes calls to get/set the time, date, system data, and process, file or device attributes.
> These system calls are used to transfer the information b/w user and the OS. Information like current time & date, no. of current users, version no. of OS, amount of free memory, disk space etc are passed from OS to the user.

v) Communication
> It includes create/delete communication connection, send/recieve messages, transfer status information, and attach/detach remote devices.
> The message passing model must support calls to:
- Identify a remote process and/or host with which to communicate
- Establish a connection b/w the two processes.
- Open and close the connection is needed.
- Transmit message along the connection.
- Wait for incoming messages, in either a blocking or non-blocking state.
- Delete the connection when no longer needed.

> The shared memory model must support calls to:
- Create and access memory that is shared amongst process.
- Free up shared memory and/or dynamically allocate it is needed

> Message passing is simple and easier.

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vi) Protection

> It provides mechanisms for controlling which users/processes have access to which system resources.

> System calls allow the access mechanisms to be adjusted as needed, and for non-privileged users to be granted elevated access permissions under carefully controlled temporary circumstances.

06) Explain the services of the operating system that are helpful for the user and the system.

→ OS provide services for the users of the system, including:

- User Interfaces - Means by which users can issue commands to the system. Depending on the O.S these may be a command-line Interface, a graphical user Interface, or a batch command systems.

> In command line Interface - commands are given to the system.

> In batch Interface - commands and directives to control these commands are put in a file and then the file is executed.

> In GUI systems - windows with pointing device to get inputs and keyboard to enter the text.

- Program Execution - The OS must be able to load a program into RAM, run the program & terminate the program, either normally or abnormally.
- I/O operations - The OS is responsible for transferring data to and from I/O devices, including keyboards, terminals, printers and files. For specific devices, special functions are provided (device drivers) by OS
- File-system Manipulation - Programs need to read and write files or directories. The services are required to create or delete files, search for a file, list the contents of a file and

change the file permissions are provided by OS

- Communications - Inter-process communications, IPC, either b/w processes running on the same processor, or b/w processes running on seperate processors or seperate machines.
- Error Detection - Both hardware and software errors must be detected and handled appropriately by the OS. Errors may occur in the CPU and memory hardware, in I/O devices, and in the user program.

OS provide services for the efficient operation of the system, inclu

- Resource Allocation - Resources like CPU cycles, main memory, storage space and I/O devices must be allocated to multiple users and multiple jobs at the same time.
- Accounting - There are services in OS to keep track of system activity and resource usage, either for billing purposes or for statistical record keeping that can be used to optimize future performance.
- Protection and security - The owners of information (file) in multiuser or networked computer system may want to control the use of that information. When several seprate processes execute concurrently, one process should not interfere with other or with OS.

07) List and Explain the different computing environments.

→ 1) Traditional Computing

> Used in office environment

- PCs connected to a network, with servers providing file & print services

> Used in home networks

- At home most users had a single computer with a slow modem.
- Some homes have firewalls to protect their networks from security breaches

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→ Web technology are streching the boundaries of traditional computing
• Network computers are terminals that understand web computing
→ Systems were either batch or interactive.
• Interactive systems waited for input from users.
• Batch system processed jobs in bulk, with predetermined input

ii) Client-server computing
→ i) computer-server
• System provides an interface to which a client can send a request to perform an action.
• In response, the server executes the action and sends back results to the client.
ii) File-server
• System provides a file-system interface where clients can create, read and delete files.
Ex: web server that delivers files to clients running web browser

iii) Peer-to-peer computing
→ All nodes are considered peers, and each may act as either a client or a server.
Advantage
→ In a client-server system, the server is a bottleneck; but in a peer-to-peer systems, services can be provided by several nodes distributed throught the network.
→ A node must first join the network of peers.

iv) web-based computing
→ This includes PC, handheld PDA & cell phones.
→ Load balancer is a new category of devices to manage web traffic among similar servers.

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> More devices becoming networked to allow web access.
> Use of O.S like windows 95, client-side, have evolved into Linux and windows XP, which can be clients and servers.

Q) List the responsibilities of the operating system for process Management and memory Management.

--> > The O.S is responsible for the following activities in connection with process Management.

- Scheduling Process and threads on the CPU.
- Creating and deleting both user and system processes.
- Suspending and resuming processes.
- Providing Mechanisms for process synchronization.
- Providing Mechanisms for process communication.

> The O.S is responsible for the following activities in connection with memory management.

- Keeping track of which parts of memory are currently being used by user.
- Deciding which processes and data to move into and out of memory.
- Allocating and deallocating memory space as needed.